

WEEK-10 Elasticbeanstalk

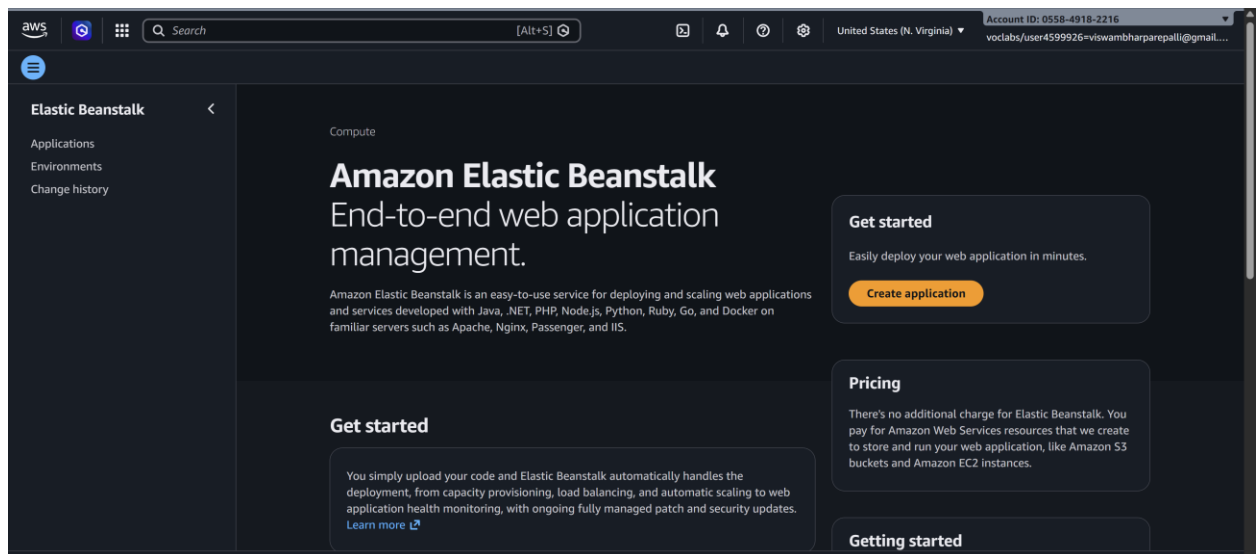
Prerequisites

- A web application ready (e.g., .war, .zip, or code)

Steps to Create Elastic Beanstalk Application

Step 1: Open Elastic Beanstalk

1. Login to AWS Console
2. Under Compute section ->Search for **Elastic Beanstalk**



- 3.
4. Click **Create Application**

Step 2: Create Application

1. Enter **Application Name** (e.g., MyApp)
2. Add description (optional)
3. Click **Create**

4.

The screenshot shows the 'Create new application' page in the AWS Elastic Beanstalk console. The left sidebar contains 'Elastic Beanstalk', 'Applications', 'Environments', and 'Change history'. The main content area is titled 'Create new application' and includes an 'Info' icon. It has two main sections: 'Application information' and 'Tags'. In 'Application information', the 'Application name' field is filled with 'MyApp' and has a note 'Maximum length of 100 characters.' The 'Description' field is empty. The 'Tags' section states 'Apply up to 50 tags. You can use tags to group and filter your resources. A tag is a key-value pair. The key must be unique within the resource and is case-sensitive. Learn more' and 'No tags associated with the resource.' There is an 'Add new tag' button and a note 'You can add up to 50 tags.' At the bottom right, there are 'Cancel' and 'Create' buttons.

Step 3: Create Environment

1. Click **Create Environment**
2. Choose:
 - **Web Server Environment** (for web apps)
3. Click **Select**

4.

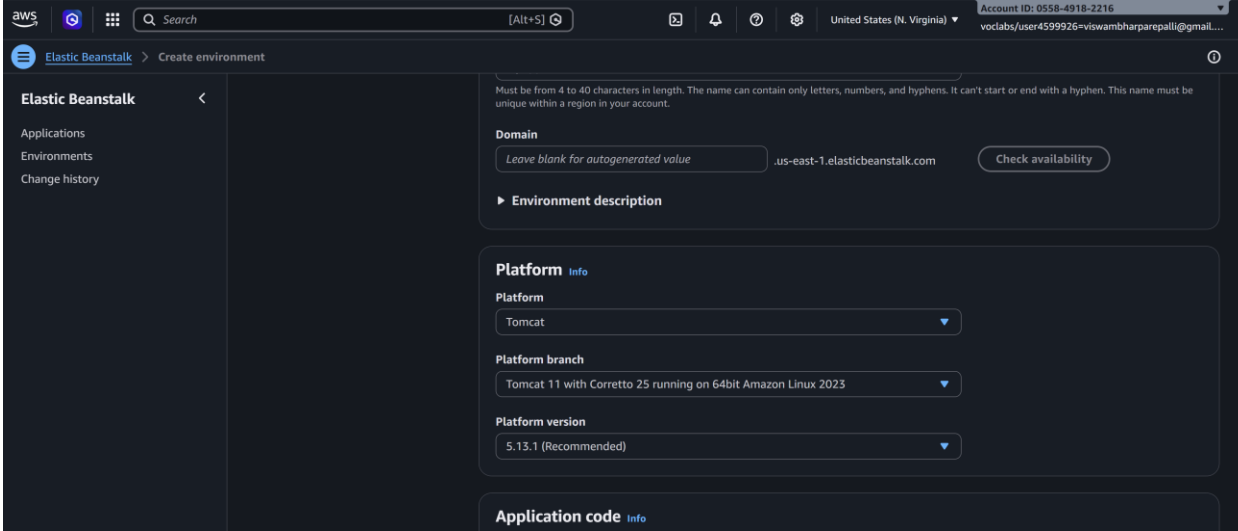
The screenshot shows the 'Configure environment' page in the AWS Elastic Beanstalk console. The left sidebar contains 'Elastic Beanstalk', 'Applications', 'Environments', and 'Change history'. The main content area is titled 'Configure environment' and includes an 'Info' icon. It has a 'Step 1' section with a list of steps: 'Configure environment' (selected), 'Configure service access', 'Set up networking, database, and tags' (optional), 'Configure instance traffic and scaling' (optional), 'Configure updates, monitoring, and logging' (optional), and 'Review'. The 'Environment tier' section has two options: 'Web server environment' (selected) and 'Worker environment'. The 'Application information' section has an 'Application name' field filled with 'MyApp' and a note 'Maximum length of 100 characters.' There is an 'Application tags (optional)' section. At the bottom, there are 'CloudShell' and 'Feedback' links, and a footer with '© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences'.

Step 4: Configure Environment

1. Enter **Environment Name**

2. Choose **Platform**:

- tomcat (for .war)
- tomcat /Node.js / Python / PHP / etc.



aws Elastic Beanstalk > Create environment

Must be from 4 to 40 characters in length. The name can contain only letters, numbers, and hyphens. It can't start or end with a hyphen. This name must be unique within a region in your account.

Domain
Leave blank for autogenerated value .us-east-1.elasticbeanstalk.com Check availability

Environment description

Platform info

Platform
Tomcat

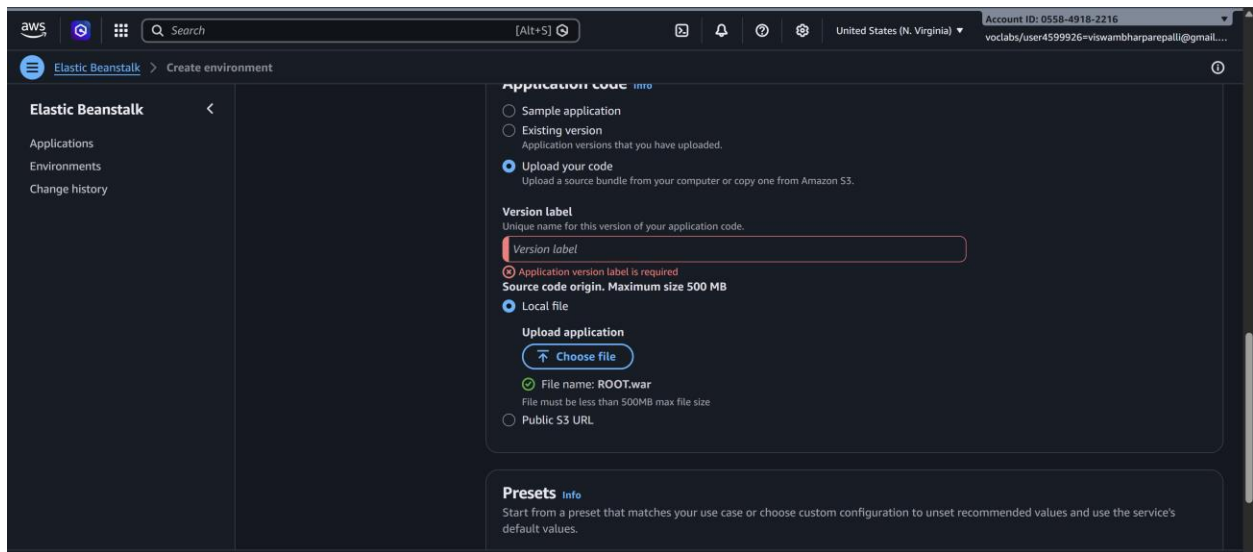
Platform branch
Tomcat 11 with Corretto 25 running on 64bit Amazon Linux 2023

Platform version
5.13.1 (Recommended)

Application code info

3. Under **Application Code**:

- Select **Upload your code**
- Upload your .war or .zip file



aws Elastic Beanstalk > Create environment

Application code info

☐ Sample application
☐ Existing version
Application versions that you have uploaded.

☒ Upload your code
Upload a source bundle from your computer or copy one from Amazon S3.

Version label
Unique name for this version of your application code.

Version label

Application version label is required

Source code origin. Maximum size 500 MB

☒ Local file

Upload application

Choose file

File name: ROOT.war
File must be less than 500MB max file size

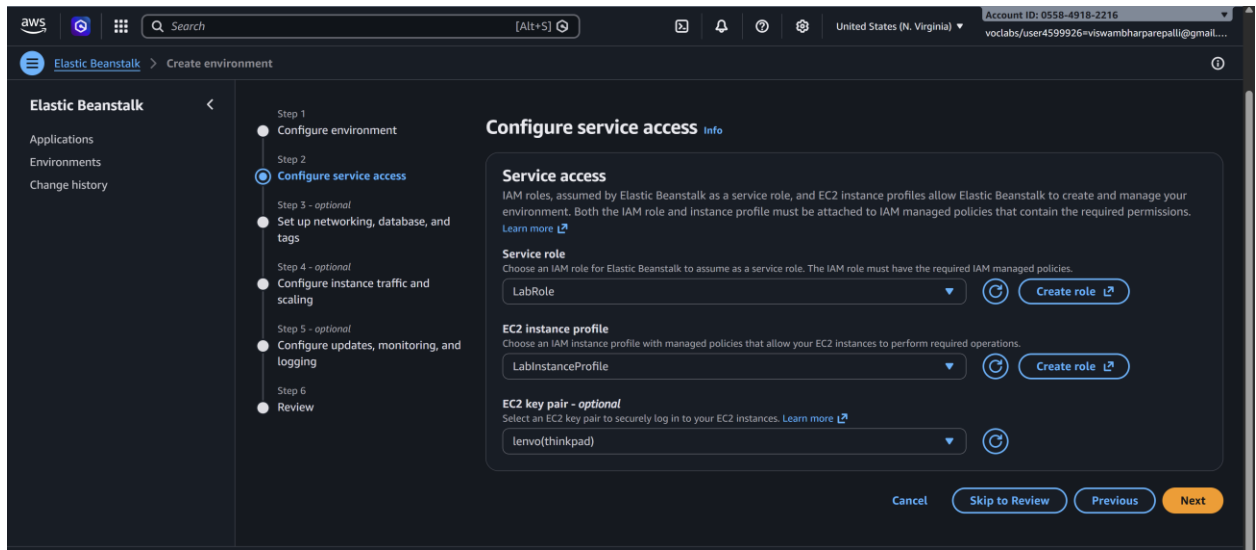
☐ Public S3 URL

Presets info
Start from a preset that matches your use case or choose custom configuration to unset recommended values and use the service's default values.

Step 5: Configure Service Access

1. Service Role → Select existing → **LabRole**
2. EC2 Key Pair → Optional (for SSH)

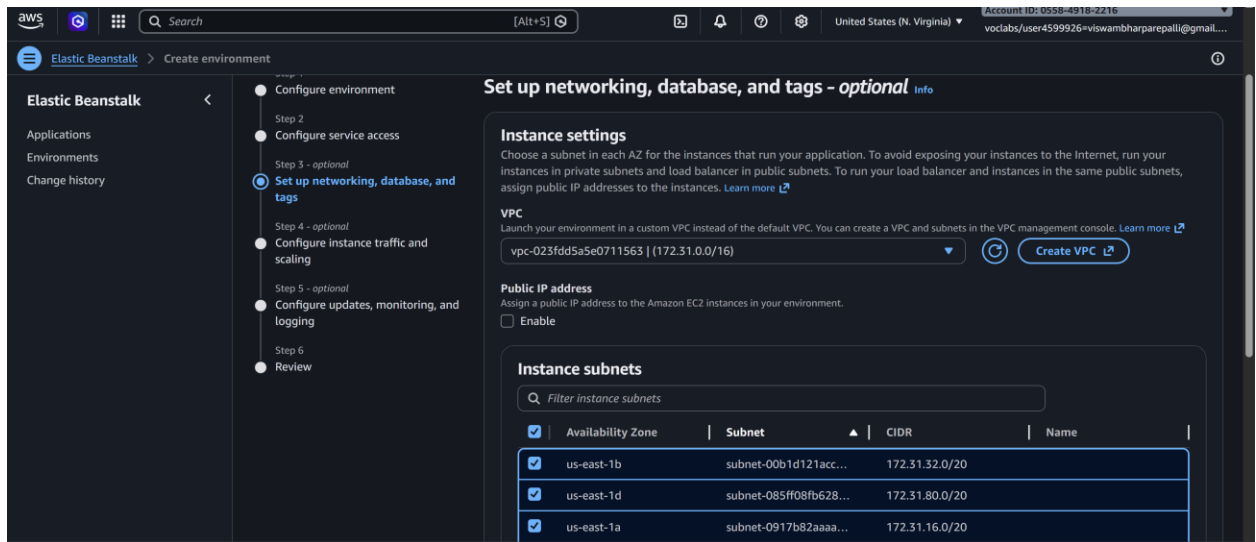
3. EC2 Instance Profile → select (e.g., LabInstanceProfile)



4.

Step 6: Configure Network

1. Select **VPC** (default is fine)
2. Select **Subnets**
3. Enable **Public IP**

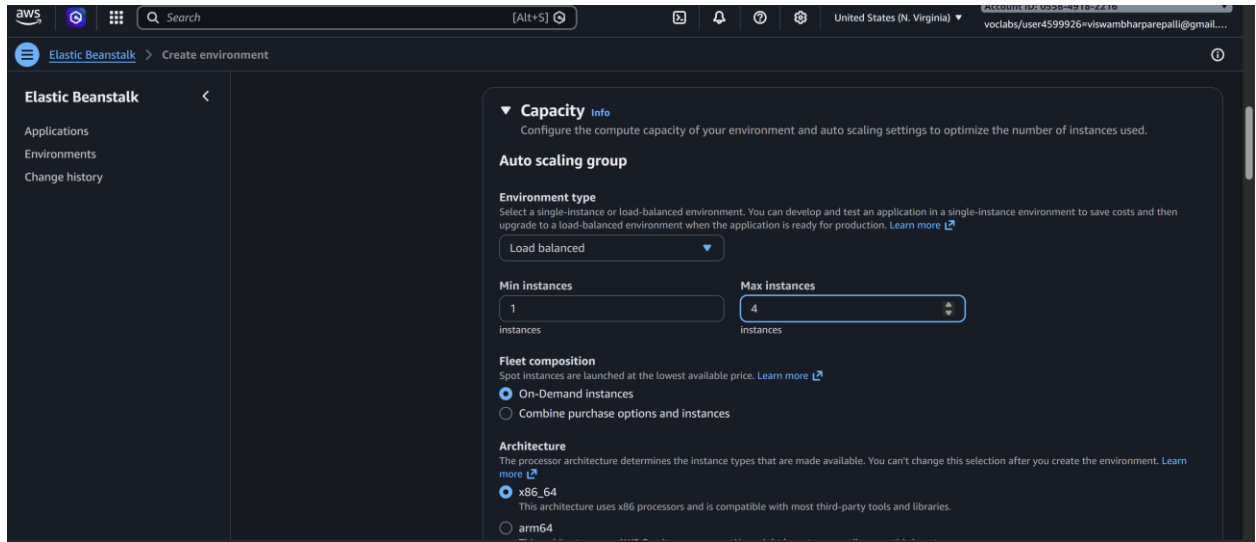


4.

Step 7: Instance & Scaling Setup

1. Choose **Instance type** (e.g., t2.micro – free tier)

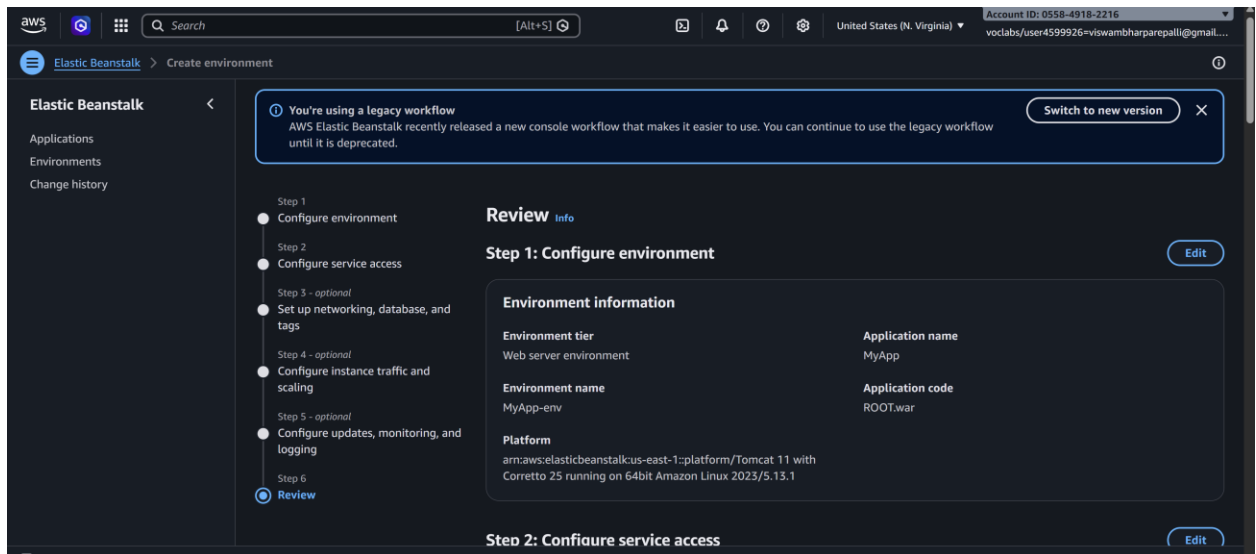
2. Set:
 - Min instances: 1
 - Max instances: 2 (or more)
3. Load Balancer → Enabled (optional but recommended)



4.

Step 8: Review & Create

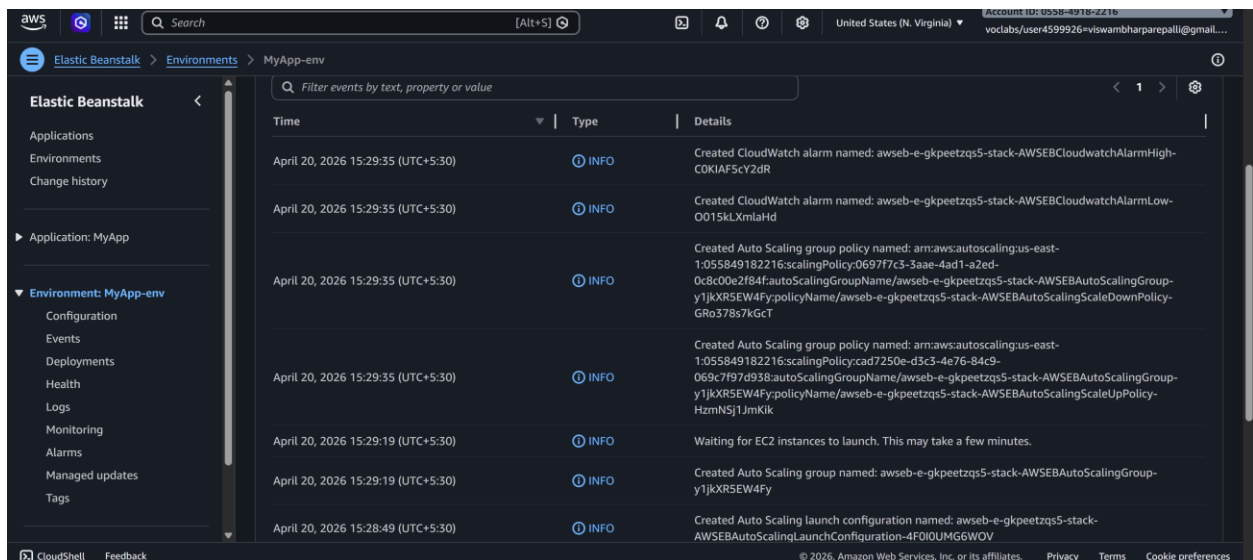
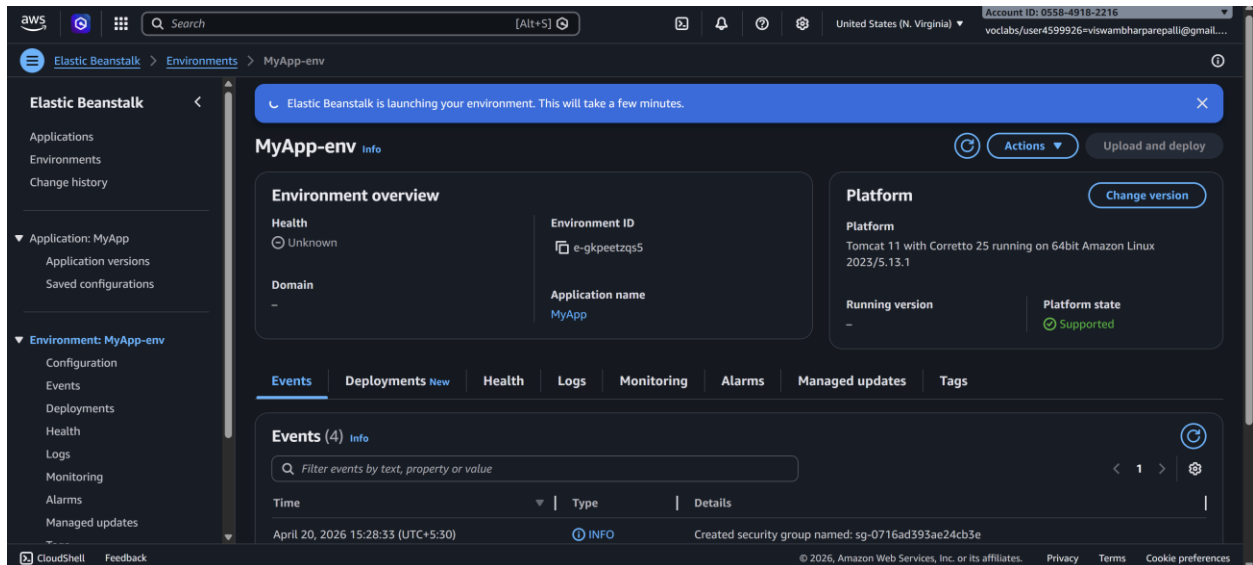
1. Click **Review**
2. Click **Create**



3.

Deployment Process

- AWS automatically creates:
 - EC2 instances
 - Load Balancer
 - Auto Scaling Group
 - Security Groups



Access Your Application

- After deployment:

- You will get a **URL (domain link)**
- Open it in browser → Your app will run

Domain

MyApp-env.eba-dp22emmf.us-east-1.elasticbeanstalk.com ↗



Sample "Hello, World" Application

This is the home page for a sample application used to illustrate the source directory organization of a web application utilizing the principles outlined in the Application Developer's Guide.

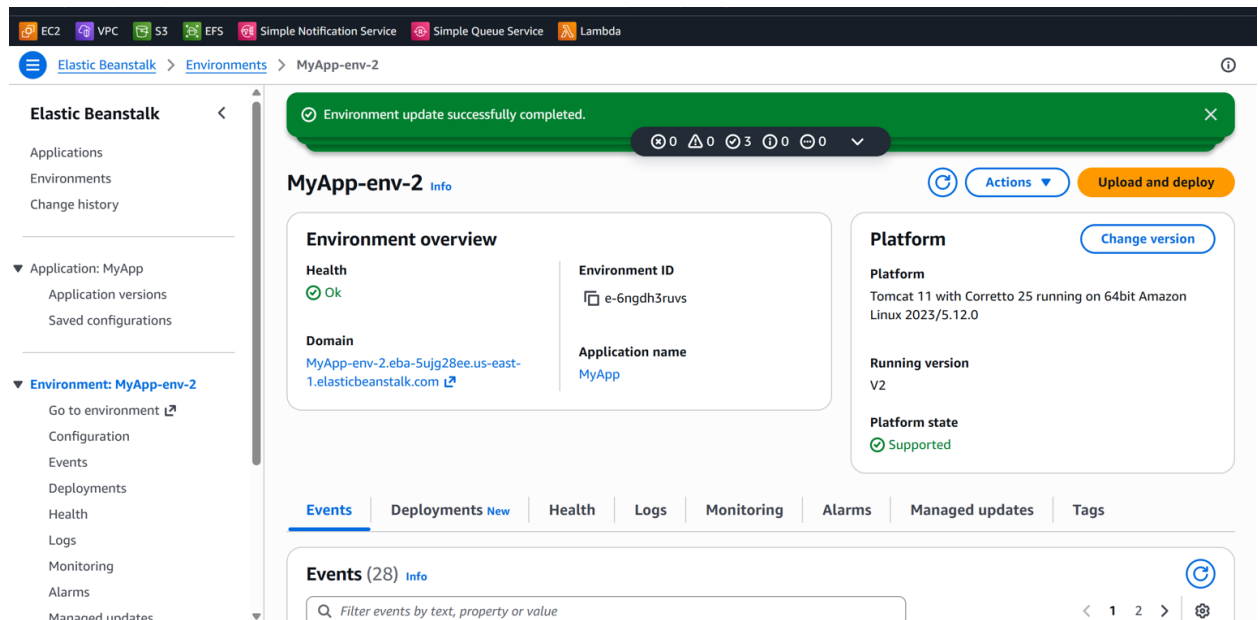
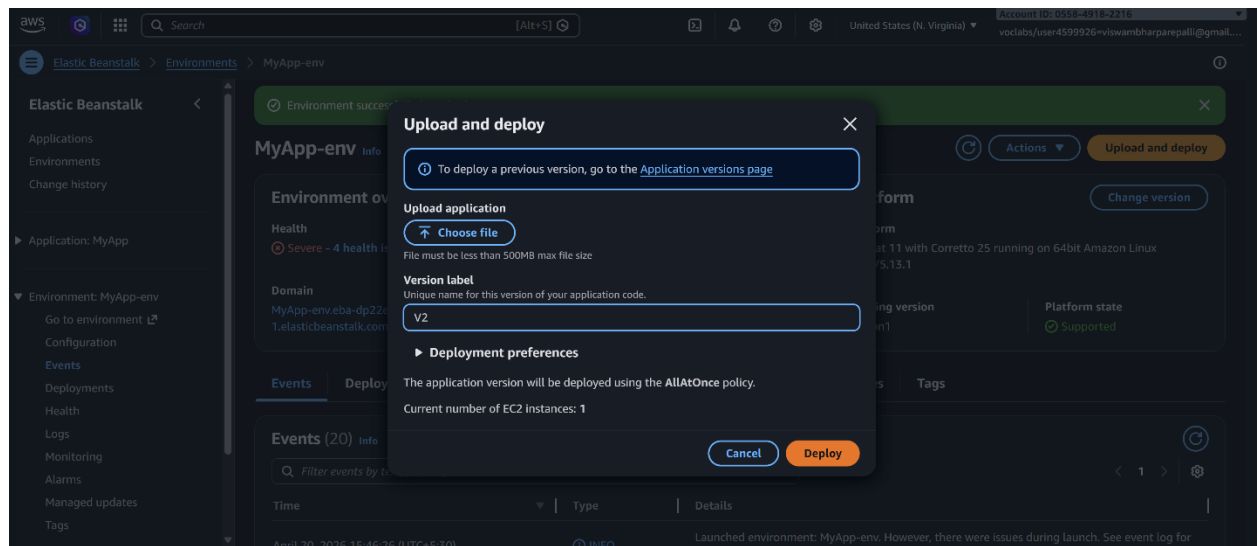
To prove that they work, you can execute either of the following links:

- To a [JSP page](#).
- To a [servlet](#).

Updating Application

1. Go to your environment
2. Click **Upload and Deploy**

3. Upload new version



Scenario-Based Questions

Scenario 1: Application Not Loading

You deployed an application using Elastic Beanstalk, but the application does not load and shows an error page.

- What could be the possible reasons for this issue?

- Could deployment or configuration be incorrect?
 - How would you troubleshoot this problem?
-

Scenario 2: Incorrect Application Structure

Your application is deployed, but the expected output is not displayed.

- How does application structure affect deployment in Elastic Beanstalk?
 - What could happen if files are placed inside an extra folder?
 - How would you fix the structure issue?
-

Scenario 3: Wrong Platform Selection

While creating the environment, an incorrect platform was selected.

- Why is platform selection important in Elastic Beanstalk?
 - What kind of errors or behavior might occur?
 - How can you correct the platform without affecting the application?
-

Scenario 4: Environment Health Warning

The environment status shows Warning (Yellow) instead of healthy.

- What are possible reasons for this warning?
 - Which sections (Events, Logs, Metrics) would you check first?
 - How can you resolve health issues?
-

Scenario 5: Application Errors in Logs

Logs show a high number of application errors.

- What do such errors indicate?
- Could this be due to misconfiguration or missing resources?

- How would you analyze logs to identify the root cause?
-

Scenario 6: Deployment Failure

The application deployment fails during upload.

- What could be wrong with the deployment package?
 - Which logs would you check to diagnose the issue?
 - How can packaging errors affect deployment?
-

Scenario 7: Works Locally but Not on Cloud

The application works correctly on a local machine but fails after deployment.

- What environment differences could cause this issue?
 - Could configuration or dependency issues be responsible?
 - How would you debug using cloud logs?
-

Scenario 8: Blank Output

The application loads but shows no output.

- What could cause an application to display no output?
 - Could runtime errors or misconfigurations be responsible?
 - How would logs help in identifying the issue?
-

Scenario 9: Application URL Not Accessible

The application URL is not accessible after deployment.

- Could the environment still be launching?
- What status indicators should be checked?
- How can Events help identify the issue?

Scenario 10: Resource Management Issue

After completing the lab, resources were not terminated.

- What are the cost implications of leaving resources running?
- Which AWS services continue running in the background?
- How can all resources be properly terminated?

Scenario 11: High Traffic Handling

The application receives sudden high traffic.

- How does Elastic Beanstalk handle scaling?
- What configurations should be enabled?
- What happens if scaling is not configured?

Scenario 12: Application Update Not Reflected

Changes are made to the application, but users still see the old version.

- Could deployment issues be the reason?
- How can caching affect the output?
- How would you ensure the latest version is visible?